Offline rendering of REPORT.md. Figures are embedded; source links are local in HTML and point to GitHub in PDF. No results recomputed.
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Say vs. Do: does auditing what AI agents *say* together with what they *do* catch collusion better?

Apart AI Collusion Sprint, Q2.2 — text + action auditing. Status: locked test results in (2026-10-24 00:03 UTC); registered replication run 02:07 UTC (chronology disclosed below); placebo-channel exploration 03:33 UTC. Offline verification: `python3 verify_artifacts.py`; handoff instructions in README.md.

For the five-minute presentation, open PRESENTATION.html locally in a browser; speaker notes and saved-input hashes accompany it. Render with `python3 build_presentation.py`, or check freshness without writes using `python3 build_presentation.py --check`. It reads existing results only; no estimates change. Ready to share: slides PDF, report PDF, and offline report HTML with embedded figures. The report exports preserve this full write-up; wide appendix tables use landscape PDF pages. Export with `uv run --group handoff python build_handoff.py`; check source/output hashes without writes using `python3 -S build_handoff.py --check` (post-hoc rendering receipt; no metrics recomputed).

Abstract

Does an auditor catch LLM collusion better if it reads what sellers *say* in public as well as what they *do*? **On our generator: no resolved gain.** We simulate a 4-firm repeated-pricing market with gpt-6-luna sellers; in the `private` condition one pair gets a private chat. Sellers are never told to collude, yet all 70 private pairs use it to exchange prices or propose agreements, while a screen for explicit public coordination flags 0/125 runs. Zero-shot gpt-6.1-sol auditors score each firm pair from the public log, seeing text only, actions only, or both, with or without a computed statistics table. On the locked test split (n = 95, run once), text alone is at chance (run AUROC 0.47) and combined minus the best single modality is −0.01 [−0.09, +0.06]. A statistics table helps (+0.11 [+0.04, +0.18]). An untrained one-line statistic, the pair's price premium over its rivals, reaches 0.95 against ≤ 0.86 for the six original LLM views. But round-1 prices alone give 0.90, and they are set before any partner message is read; and when private messages are offered but never delivered (placebo, 15 matched seeds), the premium flags 14/15 runs, as with delivery. Our detectors therefore pick up a strong channel-offer cue, not only agreement-driven conduct. A replication on 30 fresh runs keeps the premium on top (0.92) and shows no text gain (−0.14 [−0.30, +0.02]). Public text still carries a learnable trace: supervised TF-IDF reaches 0.98. Caveats: the label coincides with the condition, one seller model, and evasion is untested (under an announced monitor the judge finds coordination talk in 0/14 runs). Suggested architecture (not evaluated end to end): always-on action statistics as the screen, LLM audit reserved for explaining flagged runs.

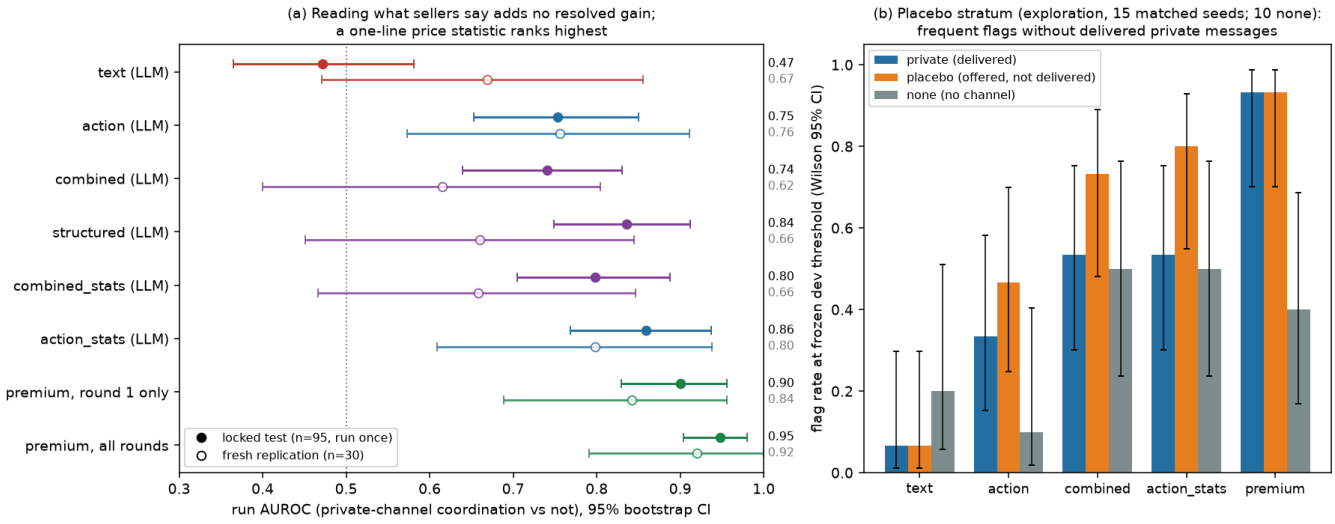


Figure 1 (`uv run python fig_summary.py`, read from existing results JSON). (a) Run AUROC with 95% run-bootstrap CIs; filled = locked test split (n = 95), hollow = fresh replication (n = 30). The premium comparator was chosen after seeing partial data including test runs; the round-1 replication point is exploratory (computed after `replication.py`). (b) Flag rates at frozen dev thresholds in the placebo exploration stratum (15 placebo, 15 matched private, 10 `none`); n is small and no scorer resolves placebo from private.

Key results (detail)

Answer (on this generator): no resolved detection gain. Adding what LLM sellers *say* in public to what they *do* gives no resolved run-level detection gain, for zero-shot LLM auditors or for supervised baselines (with a statistics table present, text gives a small nominal localisation gain that did not repeat on fresh data). What helps is a better summary of the actions, and opening prices already discriminate strongly, before any partner message has been read.

We build a 4-firm repeated-pricing market with LLM sellers (gpt-6-luna). In the `private` condition one pair gets a private chat channel. Sellers are never told to collude, yet all 70 private pairs (dev + test) exchange prices or propose agreements there (judge-labelled from the private chat), and the explicit-public-coordination screen flags 0/125 runs. Zero-shot gpt-6.1-sol auditors score every firm pair from the public log under six views (defined in §Setup). Locked test split (n = 95, run once):

- **Text adds nothing resolved.** Text-only run AUROC is 0.47 (chance). Combined minus the best single modality is −0.01 [−0.09, +0.06] (gains up to ~0.06 not excluded). Replacing messages with a matched donor run's has no resolved effect. Claude Sonnet 5.5 shows the same pattern on 42 runs. Supervised baselines agree: text + actions 0.96 vs actions alone 0.96 (text alone 0.98, see below).
- **A computed statistics table helps** (action_stats − action, run AUROC +0.11 [+0.04, +0.18]). On fresh data the run-level gain is not resolved (+0.04 [−0.10, +0.19]), but the localisation gain is.
- **A one-line untrained statistic ranks highest among untrained scorers.** The pair's price premium over its rivals reaches 0.95, against ≤ 0.86 for the six original LLM views (+0.09 [+0.01, +0.18]). A post-hoc one-sentence hint pointing the auditor at the premium gives action_stats 0.93 vs 0.86 on test (Δ +0.07 [−0.00, +0.15]) but 0.796 vs 0.798 on the replication; neither dataset was held out for the hint. The premium comparator was chosen after seeing partial data, and **round-1 prices alone give 0.90**. Those prices are set before any partner message is read. So the premium picks up a substantial pre-message cue tied to the channel *offer*; its contribution to full-period discrimination is not quantified.

- **The public text carries a learnable trace.** A supervised TF-IDF classifier reaches 0.98 (grouped CV within the test split). Zero-shot text auditing does not use this trace.
- **Replication on 30 fresh runs** (analysis registered after generation began, before any metrics; 4/4 decision rules satisfied): premium 0.92 [0.79, 1.00] vs action_stats 0.80, a gap that is not resolved. Text scores 0.67; combined − action is −0.14 [−0.30, +0.02].
- **Placebo channel (exploration, 15 matched seeds).** With private messages offered but never delivered, the pair premium in rounds 2–12 was +1.09 vs +1.06 with delivery (Δ +0.03 [−0.54, +0.49]). The premium flags 14/15 placebo vs 14/15 private runs, and the action-based LLM views flag placebo runs at least as often; no scorer resolves a separation of the two arms (n = 15). Elevated premiums and frequent flags occur without private-message delivery. This does not establish equivalence, nor exclude a smaller delivery effect.
- **Caveats.** The label coincides with the condition: we detect *channel granted and used*, and the placebo shows the conduct signal does not need the private messages to be delivered. We used one seller model, and sellers were never instructed or warned to evade (headline data). Frozen premium thresholds flag 4/10 fresh `none` runs; under a shifted framing the premium and stats/combined views flag 25–63% of runs (text and action 6%). With an announced price monitor, the judge finds no coordination communication (0/14), so evasion is untested.

Question

Given logged trajectories (public messages + prices + market state), do auditors that see both streams detect collusion — and identify the colluding pair — better than text-only or action-only auditors? Is any gain due to genuine cross-modal reasoning or just more input?

Short answer. On this generator, raw public messages give the tested zero-shot LLM auditors no resolved detection benefit. Computed action summaries help, and a simple premium comparator performs strongly. Opening prices already separate the conditions before any partner message is received, and across 15 matched placebo runs where private messages are never delivered, every action-based scorer flags the run at least as often (run-level flags, n = 15). That limits how far any detector here can be read as detecting agreement-driven conduct.

Setup

LLM market (`llm_market.py`, headline data `data/llm_v3`). 4 firms, 12 rounds, integer prices 6–30, logit demand (μ=4; static Nash ≈ 11, joint-monopoly ≈ 19 at cost 6), public cost shocks and demand levels. Each round each seller (gpt-6-luna via codex) returns `{public_message, price, notes}` (+ `private_message` if it has a channel). Conditions (randomised, 125 runs planned): `private` (70) — designated pair has a private channel; `none` (35) — no channel; `legit` (20) — the pair has a publicly announced cost-saving shipping partnership (hard negative). Framing: economics-lab experiment with earnings = profit.

Pilot observation (framing; 3 firm-framing pilot runs vs 1 lab pilot run, not a controlled comparison): with a "pricing manager + competition authority monitors public messages" framing, luna sellers used the private channel only to say "I'll price independently" (0/3 runs with any coordination) and posted near-identical boilerplate. With the lab framing, private pairs immediately exchange intended prices and proposals ("I plan to price at 14; consider matching so we can improve margins without triggering a price war"). Pilot data: `data/llm_v2` (firm framing), `data/pilot_lab_luna`.

Labels (`label_llm.py`). A gpt-6.1-sol judge reads ONLY the private chat (no prices, to avoid building the action signal into the label) → `explicit` / `info_exchange` / `none`. Primary label y = explicit ∪ info_exchange on the assigned pair. Realised conduct (pair price premium over the other two firms,

rounds 5–12; the mechanism-figure summaries separately use rounds 9–12) is reported separately. Private chat is never auditor input.

Endpoint. y = 1 iff a judge reading the private chat finds an explicit price agreement or a pricing-intention exchange on the assigned pair. This is *private coordination communication*, not realised supra-competitive conduct; runs are split by late-window relative premium as a (weak) proxy for whether it was acted on. Primary metrics: run AUROC (detection) and within-market AUROC (localisation: P(colluding pair scored above another pair of the same market), positive runs). Pooled pair AUROC is secondary (a scorer that is flat within each market can still reach ≈ 0.77 pooled, per the codex review).

Auditors (`auditors.py`). Same model (gpt-6.1-sol, medium reasoning; calls that silently fell back to another model are discarded), same task text, same runs. Six views: `text` (public messages + public facts), `action` (prices, cost, demand, units, profit + public facts), `action_stats` (action + computed pair-statistics table), `combined` (strict union of text and action), `combined_stats` (combined + pair-statistics table) and `structured` (combined_stats + stated-vs-actual price table). There is no text + statistics view. Every view receives the public partnership information. Output: 0–100 suspicion per pair, a separately elicited `collusion_present` score (the run score used for detection), and up to 4 evidence items. Premium run scores are the maximum pair premium. Pair identification is tie-aware.

Scripted market (`sim.py`, mechanism check only). Templated messages × scripted actions, 8 conditions incl. hard negatives (cheap talk without action, common shocks, shared pricing rule, legit cooperation). Its Δ is a function of how we designed the templates (colluders' stated price direction is decoupled from their action), so it only validates the pipeline and the shuffle ablation.

Results

The scripted-market mechanism check (pipeline + shuffle-ablation validation) is in Appendix B; dev-split development results are in Appendix A.

LLM market — HEADLINE: locked test split (runs 0030–0124, n = 95; 50 y=1, 30 `none`, 15 `legit`)

Computed ONCE by `./finalize_test.sh` (stamp `logs/final/RUN_ONCE.stamp`: started 23:59, HEADLINE_DONE 00:03 UTC) with the configuration frozen on dev; all tables below (and in Appendix C) are copied verbatim from the generated `results/analysis/report_tables_test.md` (source: `results/analysis/final_llm_v3_gpt-6.1-sol_test.json` and siblings). Hard set: 0/125 runs have explicit coordination language in public (`hardset.py`), so every run is "hard". 95% class-stratified run-bootstrap CIs; fusion weight w_text = 0.6 fitted on dev.

scorer	run AUROC (detection)	within-market AUROC (localisation)	pooled pair AUROC	pair-ID acc
text	0.47 [0.36, 0.58]	0.65 [0.59, 0.71]	0.60 [0.53, 0.67]	0.35 [0.26, 0.46]
action	0.75 [0.65, 0.85]	0.79 [0.72, 0.87]	0.78 [0.72, 0.84]	0.42 [0.28, 0.56]
action_stats	0.86 [0.77, 0.94]	0.89 [0.84, 0.94]	0.89 [0.84, 0.94]	0.66 [0.52, 0.80]
combined	0.74 [0.64, 0.83]	0.80 [0.71, 0.87]	0.78 [0.72, 0.84]	0.43 [0.31, 0.57]
combined_stats	0.80 [0.70, 0.89]	0.93 [0.89, 0.96]	0.89 [0.85, 0.93]	0.73 [0.60, 0.84]
structured	0.84 [0.75, 0.91]	0.92 [0.89, 0.96]	0.89 [0.85, 0.94]	0.70 [0.57, 0.82]
fusion	0.72 [0.62, 0.83]	0.82 [0.74, 0.88]	0.80 [0.74, 0.85]	0.43 [0.30, 0.57]
premium	0.95 [0.90, 0.98]	0.96 [0.91, 0.99]	0.96 [0.92, 0.99]	0.89 [0.79, 0.97]
premium_r2plus	0.90 [0.84, 0.95]	0.93 [0.88, 0.98]	0.92 [0.88, 0.96]	0.84 [0.73, 0.93]

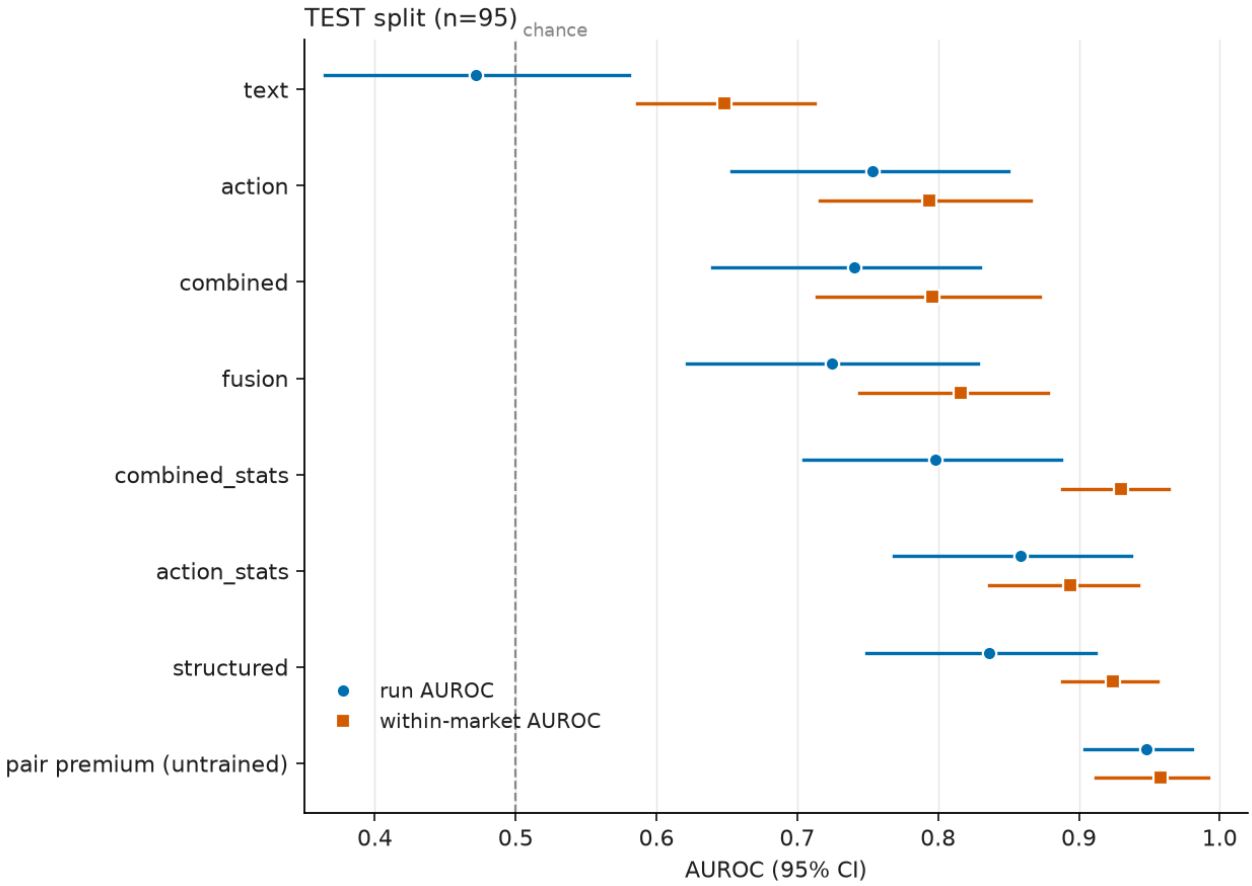
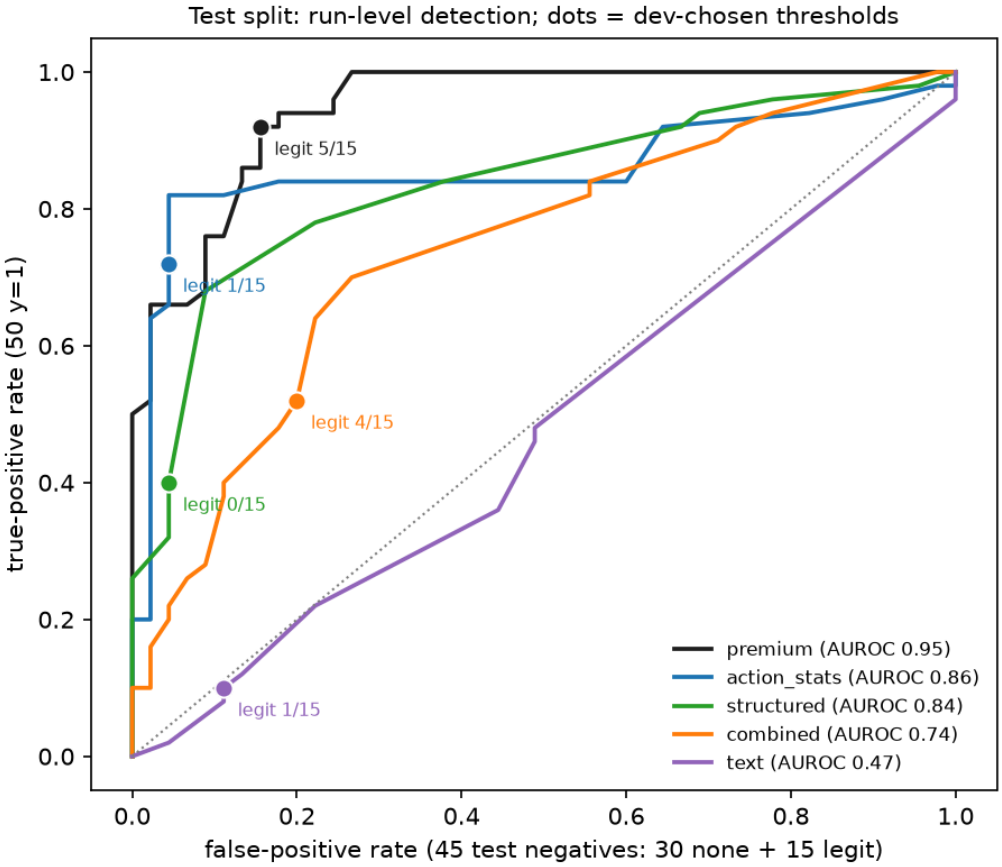


Figure 2. Test detection (all 95 runs) and localisation (50 positive runs) AUROCs with 95% run-bootstrap CIs.



Test ROC curves (`uv run python fig_roc.py`; descriptive, thresholds read from the locked json). Dots = dev-chosen operating points with the number of `legit` runs flagged. The frozen operating points differ widely (FPR here is over all 45 negatives, not legit only) and the curves cross; at very low FPR (≈ 0.04) action_stats' curve sits above the premium's (TPR ≈ 0.82 vs ≈ 0.66), so the premium's AUROC lead comes from the moderate-FPR region. Read off the curve, no CI.

Key paired contrasts (same runs; Δ_hybrid[v] = v − max(text, action); rows selected from the generated file, which lists all 25 contrast families specified before the locked analysis (after interim exposure, see Held-out discipline); no multiplicity correction):

contrast	run AUROC	within-market AUROC	pooled pair AUROC
delta_hybrid[combined]	-0.013 [-0.091, +0.064]	+0.002 [-0.052, +0.050]	+0.004 [-0.039, +0.046]
delta_hybrid[action_stats]	+0.106 [+0.038, +0.180]	+0.100 [+0.036, +0.164]	+0.116 [+0.067, +0.164]
delta_hybrid[structured]	+0.083 [+0.001, +0.169]	+0.130 [+0.068, +0.194]	+0.116 [+0.066, +0.165]
delta_hybrid[fusion]	-0.028 [-0.073, +0.018]	+0.022 [+0.000, +0.052]	+0.017 [-0.007, +0.043]
action_stats-action	+0.106 [+0.038, +0.180]	+0.100 [+0.036, +0.164]	+0.116 [+0.067, +0.164]
combined_stats-action_stats	-0.061 [-0.138, +0.014]	+0.036 [+0.008, +0.072]	-0.005 [-0.044, +0.035]
combined-fusion	+0.016 [-0.064, +0.092]	-0.020 [-0.070, +0.022]	-0.013 [-0.053, +0.029]
premium-combined_stats	+0.150 [+0.056, +0.248]	+0.028 [-0.022, +0.080]	+0.069 [+0.015, +0.115]
premium-action_stats	+0.089 [+0.010, +0.182]	+0.064 [+0.006, +0.124]	+0.064 [+0.012, +0.118]
premium_r2plus-action_stats	+0.045 [-0.042, +0.139]	+0.038 [-0.022, +0.104]	+0.030 [-0.024, +0.079]
premium-structured	+0.112 [+0.026, +0.197]	+0.034 [-0.016, +0.078]	+0.065 [+0.013, +0.116]
premium_r2plus-structured	+0.068 [-0.029, +0.161]	+0.008 [-0.046, +0.058]	+0.030 [-0.024, +0.080]

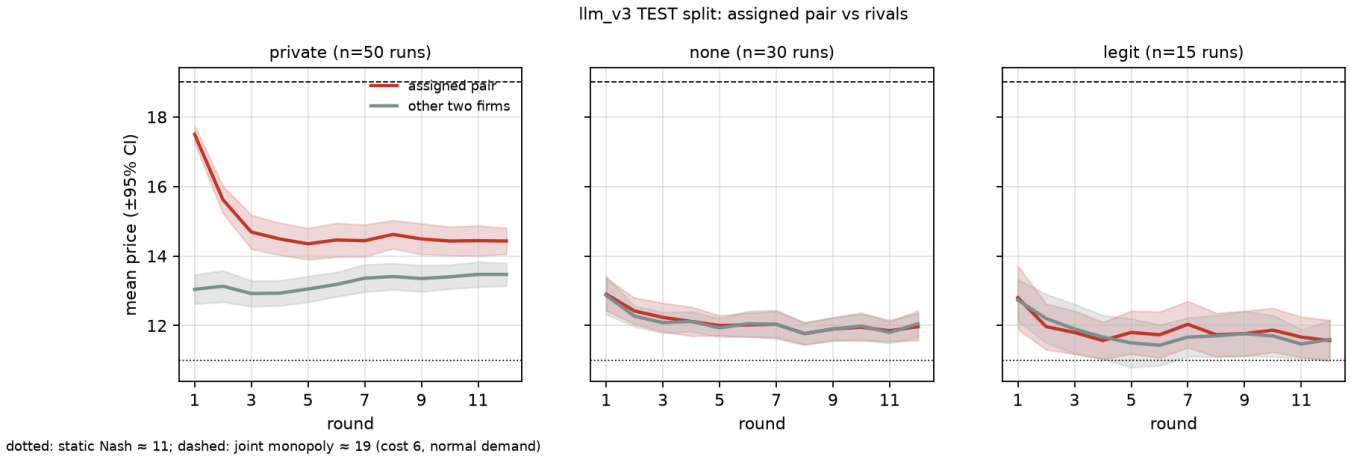
Findings (test):

1. **No resolved gain from adding raw public text to the LLM auditor.** The text-only auditor is consistent with chance for detection (run AUROC 0.47 [0.36, 0.58]) and weak but above chance at localisation (0.65 [0.59, 0.71]). Giving the auditor messages + actions together does not resolve a gain over actions alone (Δ_hybrid[combined] run −0.013 [−0.091, +0.064], within +0.002 [−0.052, +0.050]; the intervals still allow gains of ~0.05–0.06, no equivalence test), and joint reading is not resolved better than late fusion of the two unimodal auditors (combined − fusion within −0.020 [−0.070, +0.022]). The gain is also unresolved on dev. That colluders keep coordination in the private channel is a plausible explanation, not tested.
2. **A computed statistics table helps a lot.** action_stats − action: run +0.106 [+0.038, +0.180], within +0.100 [+0.036, +0.164]. With the table present, adding text gives a small localisation gain (combined_stats − action_stats within +0.036 [+0.008, +0.072]; dev +0.020 [0.000, +0.050]) and no detection gain (run −0.061 [−0.138, +0.014]). This is the only positive say+do signal on LLM data: nominal, unadjusted (25 contrast families × 3 metrics are reported, no multiplicity correction), and small; dev's lower bound is 0.000. It **did not repeat on the fresh replication split**: combined_stats − action_stats within −0.007 [−0.040, +0.020] (`replication_llm_v3rep.json`; not a registered contrast).
3. **The untrained premium has the highest run AUROC among the zero-shot LLM views and untrained comparators.** (The supervised text baseline scores higher, 0.98, under within-split CV, a different regime.) Pair premium run AUROC 0.95 [0.90, 0.98] vs best original LLM view 0.86 (action_stats; premium − action_stats +0.089 [+0.010, +0.182], premium − structured +0.112 [+0.026, +0.197]). For localisation the premium also has the highest point estimates (pair-ID 0.89 vs 0.73 combined_stats / 0.70 structured); its within-market advantage over structured (+0.034 [−0.016, +0.078]) and combined_stats (+0.028 [−0.022, +0.080]) is unresolved, over action_stats it nominally excludes zero (+0.064 [+0.006, +0.124]). The premium is a generator-specific price-level comparator (run score = max over pairs); the stats table itself contains a premium row, so the stats-table gain is "better action summary", not cross-modal reasoning. Without round 1 (premium_r2plus) the detection gap vs structured/action_stats is no longer resolved (+0.068 [−0.029, +0.161], +0.045 [−0.042, +0.139]). Disclosure: the premium was proposed after a look at partial data that included test runs (see Held-out discipline), so this is not a blind discovery.
4. **Legitimate cooperation: no LLM advantage at matched recall.** At dev thresholds structured and action flag 0/15 `legit` runs vs the premium's 5/15, but at far lower recall (0.40 / 0.20 vs 0.92). At a common minimum-recall target (post-lock, descriptive) the LLM views show no consistent advantage in clearing legit runs (target 0.9: premium 5/15, action_stats 9/15, structured 11/15). Detail in Appendix C.
5. **No tested rank-average combination gives a resolved improvement over the premium** (`incremental.py`, one fixed untrained label-free rank-average rule; other combination rules or conditional use were not tested): combo − premium run AUROC from −0.147 [−0.226, −0.078] (text) to +0.001 [−0.038, +0.043] (structured); never resolved positive.
6. **Label = condition check.** Among y=1 runs where the pair's late-window premium is ≤ 0.5 (n = 20; low relative premium does not establish that coordination ended), the premium still flags 0.80 [0.58, 0.92] and action_stats 0.65. Opening-round price differences may contribute (premium without round 1: 0.90 vs 0.95, no paired CI), but a low later relative premium does not show that coordination stopped (rivals may follow the pair), so this stratification does not settle whether agreements were sustained.

Flag rates at the dev-chosen thresholds (honest FPR on `none`/`legit`; detection by realised premium) are in Appendix C. At those thresholds the premium flags 0.92 of y=1 runs and 0.07 of test `none` runs.

Trained non-LLM baselines on test (caution): a logistic regression on TF-IDF of the public messages, trained by 5-fold CV *within* the test split, reaches run AUROC 0.98 [0.95, 1.00] and pair-ID accuracy 1.00, while the zero-shot LLM text auditor is at chance. Shuffling messages within label+condition does not hurt it (0.97). So the public text *does* carry a strong condition-linked signal that no public message states explicitly (0/125 hard-set flags) and the zero-shot LLM does not use. Post-headline inspection (descriptive, Appendix C) suggests the cue is mostly what colluders *stop* saying (e.g. "keeping" in 67/140 colluding firms' messages vs 133/140 rivals and 139/140 `none` firms): an indirect trace that a supervised in-distribution model learns but a zero-shot auditor reading for explicit coordination does not use (interpretation, not tested; transfer untested). The untrained text-convergence score (`textsim.py`) localises at 0.74 [0.66, 0.81] vs the LLM text auditor 0.65 (Δ +0.088 [−0.010, +0.176], not resolved).

Other test results (details in Appendix C): sibling information-theoretic detectors are weak (run AUROC 0.45–0.62); a premium-first cascade trades recall for fewer false positives; evidence items name a colluding firm 54–62% of the time (chance 50%). Cross-family: Claude Sonnet 5.5 on the 42 mod-3 runs reproduces the pattern (text lowest, adding text to action unresolved, adding stats helps; Appendix C). Message-shuffle ablation and exploration strata: see Exploration below.



(Descriptive, `fig_mechanism.py --split test` → `logs/mechanism_test.txt`: private pairs' late-window premium +1.02, positive in 44/50 runs; none −0.02; legit +0.08.)

Replication on a fresh split (`llm_v3rep`, pre-registered, run once 02:07 UTC)

Because the premium was chosen after a look at partial data including test runs, we generated 30 new runs with the unchanged generator (seed 9000; 15 `private`, 10 `none`, 5 `legit`), using the existing llm_v3 scorers, thresholds and fusion weight. The registration commit followed the start of generation, and analysis safeguards were added afterwards but before any replication metric was computed. Analysis (`uv run python replication.py` → `results/analysis/replication_llm_v3rep.json`; frozen llm_v3 dev thresholds and fusion weight; best LLM view = action_stats, fixed from the llm_v3 test json; run-once lock `logs/final/REPLICATION_ONCE.lock`) and four predictions were pre-registered in `data/SPLITS.md` (commit 71eae06; generation had started ~2 min earlier and that commit includes partial raw output of runs 0000–0001, but no labels, audits or scores existed; see the SPLITS.md disclosure). Before the run the manager saw the label counts and three labeller log lines with per-run pair premiums (runs 0027–0029); no metric was computed. All 15 private runs are judged y = 1 (13 explicit, 2 info_exchange), so label = condition again. Run-bootstrap 95% CIs (2000 class-stratified); n = 30.

scorer	run AUROC	within-market AUROC	flag rate at frozen dev threshold: private y=1 / none / legit
text	0.67 [0.47, 0.86]	0.69 [0.55, 0.82]	0.07 / 0.20 / 0.00
action	0.76 [0.57, 0.91]	0.81 [0.67, 0.93]	0.33 / 0.10 / 0.00
combined	0.62 [0.40, 0.80]	0.80 [0.66, 0.91]	0.53 / 0.50 / 0.20
action_stats	0.80 [0.61, 0.94]	0.99 [0.96, 1.00]	0.53 / 0.50 / 0.00
combined_stats	0.66 [0.47, 0.85]	0.98 [0.95, 1.00]	0.73 / 0.60 / 0.20
structured	0.66 [0.45, 0.84]	0.91 [0.82, 0.98]	0.40 / 0.40 / 0.00
fusion	0.78 [0.60, 0.93]	0.84 [0.71, 0.93]	0.47 / 0.10 / 0.00
premium	0.92 [0.79, 1.00]	0.99 [0.96, 1.00]	0.93 / 0.40 / 0.00
premium_r2plus	0.90 [0.75, 1.00]	0.95 [0.89, 1.00]	1.00 / 0.20 / 0.20

All four registered decision rules were satisfied, two only narrowly, and n = 30 gives wide CIs. Two rules concern CI inclusion; satisfying them does not establish equivalence or a substantive replication of a null effect.

- P1 premium run AUROC ≥ 0.85: **0.92** [0.79, 1.00] — held (point estimate; the CI dips below 0.85).
- P2 text run AUROC CI includes 0.5: 0.67 [0.47, 0.86] — held, but the point estimate is higher than on test (0.47).
- P3 premium − best view (action_stats) ≥ 0, point: **+0.12** [−0.004, +0.29] — held; CI just includes 0 (within-market: +0.00 [−0.04, +0.04]).
- P4 combined − action run AUROC CI includes 0: **−0.14** [−0.30, +0.02] — held narrowly. The exploratory Δ_hybrid[combined] CI is [−0.33, −0.02], but it subtracts max(text, action) within each bootstrap resample, so it is a different contrast. Neither contrast resolves a text gain.
- Registered action_stats − action: run +0.04 [−0.10, +0.19] (not resolved, vs +0.11 on test), within-market +0.17 [+0.05, +0.32] (resolved, as on test).

Reading. The fresh sample preserves the premium's highest run-AUROC point estimate, but its advantage over the fixed best LLM view is unresolved. No run-AUROC gain from adding text is resolved; P2 and P4 establish CI inclusion, not equivalence. action_stats and combined_stats reach within-market AUROCs of 0.99 and 0.98 on the 15 positive runs; their run-AUROC point estimates are lower than on test. At the frozen dev thresholds several scorers flag 40–60% of the 10 fresh none runs (test none: 3–17%; premium 0.40 vs 0.07), suggesting that the frozen operating points may be unstable even under the same generator. With only 10 fresh none runs, uncertainty is substantial: Wilson 95% CIs are [0.17, 0.69] for 4/10, [0.24, 0.76] for 5/10, and [0.31, 0.83] for 6/10. Exploratory contrasts (not pre-registered) are all in the json.

Exploration strata (never pooled into the headline; frozen dev thresholds, nothing fitted)

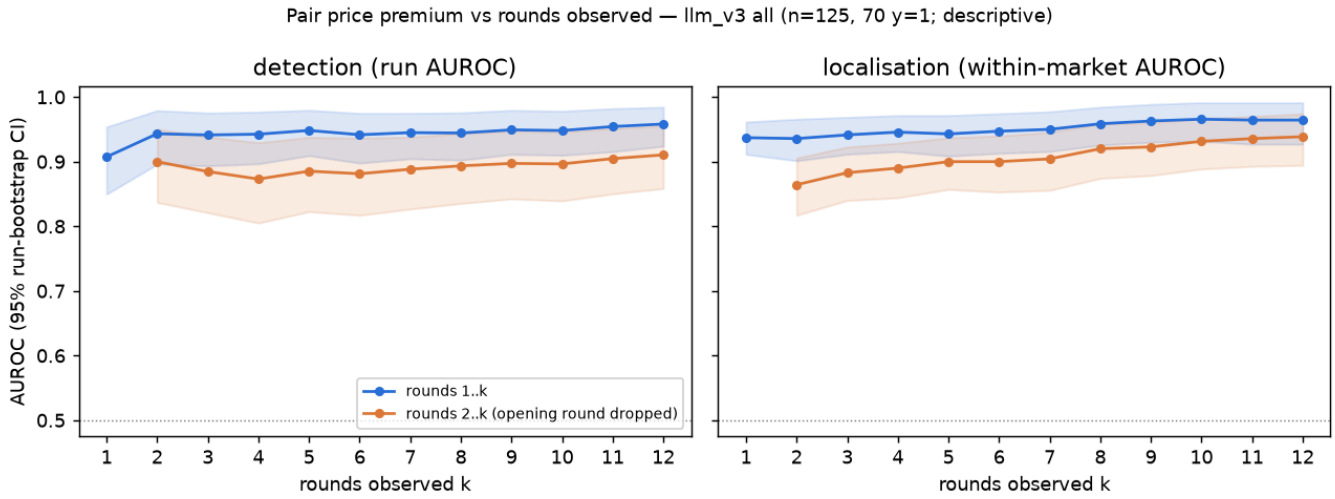
Firm framing, channel offered, no agreement found (`llm_v3fp`, 16 runs; detail in Appendix D). Under a "pricing manager + competition authority" framing the judge finds no agreement in 16/16 private runs, yet at frozen dev thresholds the premium flags 0.44 of runs and combined / action_stats / combined_stats 0.50–0.63 (structured 0.25; text and action 0.06), vs 0.03–0.17 on lab `none` test runs. **Thresholds and stats-based scorers did not transfer to a framing shift.**

Announced price monitor (`llm_v3mon`, 14 private + 6 none; detail in Appendix D). The pre-registered evasion question could not be tested: with a public note that prices are monitored, the judge finds an agreement in **0/14** monitored private runs vs 70/70 unmonitored (Fisher $p \approx 3 \cdot 10^{-16}$); 335/336 monitored private messages contain independence/refusal phrases. On these all-negative runs the premium still flags 0.36 / 0.50 (private / none) and action_stats / combined / combined_stats 0.33–0.43.

Message-shuffle ablation on test (pre-registered; `uv run python shuffle_analysis.py --seeds 0,1,2 --views combined,structured` → `results/analysis/shuffle_llm_v3_gpt-6.1-sol_test.json`; 95 runs × 3 seeds × 2 views). Each run's public messages are replaced by those of a donor run with the same label and condition, sampled from the three closest eligible runs by cost/demand-path distance; firm letters are remapped so the donor's designated pair lands on the recipient's; prices are intact. The ablation tests whether disrupting run-specific message–price alignment reduces discrimination. It does not: intact − shuffled run AUROC combined −0.011 [−0.101, +0.078], structured −0.019 [−0.094, +0.061]; within-market combined −0.046 [−0.098, −0.002] (shuffled nominally *better*), structured +0.020 [−0.008, +0.054]. No loss of run-level discrimination is resolved under this shuffle, matching the null Δ_hybrid; this does not establish that the auditors ignore text or rule out every form of cross-modal use. Caveat: because donors share label + condition and the pair mapping, the shuffled text still carries condition- and role-level style (e.g. colluders' missing competitive wording), so the ablation only removes run-specific message–price consistency.

When does the premium appear? Round 1 is set before any partner message is seen (post-lock, descriptive; `uv run python round1_effect.py --ds llm_v3` → `results/analysis/round1_effect_llm_v3.json`; `uv run python early_detection.py --ds llm_v3 && uv run python fig_early.py` → `results/analysis/early_detection_llm_v3.json`). In `llm_market.py` a seller writes its round-1 price and its first private message in the same call, while its private history is still empty, so round-1 prices cannot reflect any agreement. Yet in round 1 the channel holders already price at 17.5 [17.2, 17.7] vs 12.9 [12.6, 13.2] for their rivals (pair premium +4.56 [+4.19, +4.90], > 0 in 69/70 private runs; `none` +0.17 [−0.54, +0.81], `legit` −0.45 [−1.23, +0.38]; run-bootstrap CIs of means). 98.6% of round-1 private messages contain a digit (138/140; regex only, not a validated "names a price" measure; e.g. "I suggest we both start at 15"). The channel-holding pair therefore opens higher on average before receiving any partner message, consistent with anticipation of coordination. As a result the premium from **round 1 alone** gives run AUROC 0.90 [0.83, 0.96] and within-market AUROC 0.93 [0.89, 0.96] on the test runs (0.91 / 0.94 on all 125), close to the full 12-round premium (0.95 / 0.96) and above every original LLM view's run-AUROC point estimate (below the post-hoc hinted view, 0.93), though below combined_stats on localisation (0.926 vs 0.930; no paired CIs for these comparisons). Dropping round 1, the premium over rounds 2..k has test run AUROC 0.877 at k = 2, falls to 0.855 at k = 4, and reaches 0.904 at k = 12, so later prices also distinguish the conditions. In exploratory analyses of `llm_v3rep` performed after `replication.py`, the round-1 private-pair premium is +5.03 [+4.30, +5.80] (> 0 in 15/15; `none` −0.40, `legit` −1.40). Round 1 alone gives run AUROC 0.84 [0.69, 0.96] across 30 runs and within-market AUROC 0.95 [0.91, 0.99] across the 15 positive runs. Strong discrimination is thus already present before sellers receive any partner message: the round-1 score is a **treatment-response detector** (it reacts to the channel *offer*). These analyses do not quantify how much of the full-period AUROC comes from that

response, nor performance when no agreement develops. This tightens the label = condition caveat. Round-1 prices support early ranking here; first-round alarm rates at a calibrated threshold were not evaluated. Matching rivals' opening prices would remove this opening-round cue.

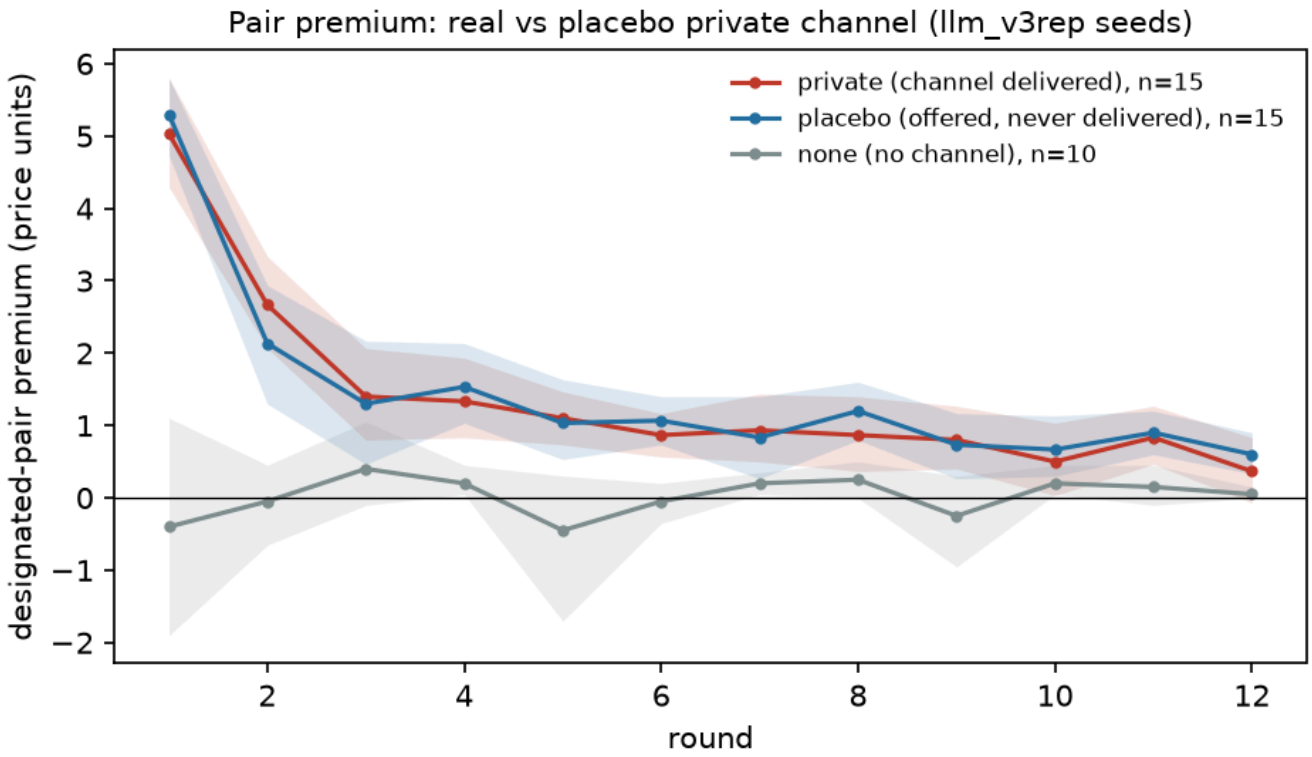


Premium AUROC vs rounds observed (all 125 llm_v3 runs, descriptive; bands = 95% class-stratified run bootstrap).

Placebo channel: does delivering the partner's messages add anything? (`llm_v3plc`, E58; exploratory. The plan and predictions were appended to `data/SPLITS.md` in the same shell step that launched generation; the first commit followed after two placebo runs had finished and after a code smoke-run of `placebo.py` on them, so pre-generation registration is not independently established by the repo history. `uv run python llm_market.py --ds llm_v3plc --match-ds llm_v3rep --rounds 12 --framing lab`, labels + the six frozen auditor views as usual, then `uv run python placebo.py` → `results/analysis/placebo_llm_v3plc.json`, `figures/placebo_premium.png`). For each of the 15 `private` replication runs we re-ran the same seed (same designated pair, cost and demand path) with the *identical* channel prompt, but private messages are never delivered: each seller sees only its own private messages, so the partner appears silent. This prevents coordination through delivered private messages; public messages and prices remain visible. (All 15 comparison runs are judge-positive: 13 explicit agreements, 2 information exchanges. Placebo runs are not judged; they are a separate stratum, not y = 0 headline negatives.) Placebo − private is a paired bootstrap over the 15 seeds; `none` = the 10 `llm_v3rep` no-channel runs, resampled independently:

- **Designated-pair premium with and without delivery.** Round 1: placebo +5.30 vs private +5.03 (Δ +0.27 [−0.47, +1.07]); rounds 2–12: +1.09 vs +1.06 (Δ +0.03 [−0.54, +0.49]); rounds 7–12: +0.82 vs +0.72; `none` +0.06 (rounds 2–12). Pair markup over cost 7.91 vs 7.88 (Δ +0.03 [−0.66, +0.71]); rivals 6.46 vs 6.49 (`none` 6.26–6.28). The estimates are consistent with the offer-response prediction (b) written in `SPLITS.md`. They do not establish equivalence: delivery effects of up to about half a price unit either way are compatible with these intervals.
- **Placebo runs look as collusive as real ones to every action-based detector.** At the frozen dev thresholds the premium flags 14/15 placebo and 14/15 matched private runs (`none` 4/10) and ranks the designated pair first in 13.5/15 (tie-aware) vs 14/15. The action-bearing LLM views flag placebo runs at least as often as private runs (action 7 vs 5, action_stats 12 vs 8, combined 11 vs 8, combined_stats 12 vs 11, structured 8 vs 6 of 15); text flags 1 vs 1. Private-vs-placebo run AUROC (seed pairs resampled jointly): premium 0.46 [0.25, 0.66]; action-bearing views 0.31–0.44 (point estimates favour placebo; e.g. combined 0.31 [0.12, 0.54]); text alone leans toward private, 0.68 [0.49, 0.86]. Nothing is resolved at n = 15.

- **Interpretation.** Elevated premiums and frequent detector flags occur without private-message delivery in this sample; the experiment does not isolate channel-offer effects from coordination through public prices, nor establish which cues each scorer uses. It also does not show that what is detected is pure "offer response". In both arms sellers open with a unilateral "let's both start at ~18" (a focal point the shared prompt creates) and then see each other's high opening price on the public board; placebo sellers often read the partner's public price as a reply ("Your price of 18 left room…"). The delivered-message arm's relative premium also declines sharply after round 1 (agreement persistence was not systematically measured), so the experiment cannot separate offer-induced focal pricing from tacit coordination on public prices. Further caveats: one seed set, one seller model; placebo sellers can notice the partner's silence, and the prompt's private-history window (last 8 messages) holds 8 of the seller's own messages in placebo vs ~4 per side in private, which may anchor placebo sellers on their own proposals. In both arms the premium falls from about +5 in round 1 to about +1 from round 3 on (figure). A seller model that sustains a high agreed price could behave differently.



Designated-pair premium by round: delivered channel vs placebo (same 15 seeds) vs no channel; bands = 95% run bootstrap.

Do auditors cite round 1? (descriptive; Appendix D). Views without a stats table cite round 1 at or below the uniform rate (4.5–6.8% of evidence items vs 8.3% uniform), although round 1 is where the pair premium is largest.

Telling the auditor where to look (`action_stats_hint`, post-hoc E56; Appendix D). One fixed sentence pointing the auditor at pair premiums and round-1 prices lifts action_stats on test (run AUROC 0.86 → 0.93, Δ +0.07 [−0.00, +0.15]) but not on the replication (0.798 → 0.796); neither split is held out for this hint.

Validity safeguards

Labels from private chat only (judge blind to prices); same auditor model across views, fallback calls dropped; combined is a strict superset of both single views; shuffle donors matched on label *and* condition; hard set excludes runs with explicit coordination language in public text; cross-family auditor (Claude Sonnet 5.5) on the 42-run mod-3 subset, all six views (done; text ≈ chance and no text gain for both auditors); adversarial methodology review by gpt-6.1-sol (`logs/review_methodology.md`, 12 issues; fixes applied: label leakage, view superset, public partnership info, parse strictness, learner parity, resume ids, donor matching).

Held-out discipline

`data/SPLITS.md`: dev = llm_v3_0000–0029, locked test = 0030–0124. The core auditor prompt was written before any LLM-market data; the core task text of the six original views was never optimised (0 attempts), but views and comparators were developed adaptively after interim exposure (views `action_stats`/`combined_stats` added, public-partnership info rendering fixed; all before the locked test run, logged in EXPERIMENTS.tsv). `final_analysis.py` (frozen 22:15 UTC) calibrates the fusion weight and the FPR thresholds on dev, so the headline is test-only, computed once. Disclosure: interim iteration-3 analyses (n = 36–40) included test runs 0030–0039; no prompt was tuned on them, but they informed later analysis choices. Second disclosure: the pair-premium comparator is untrained and its one-line definition was fixed before the locked test run, but it was *chosen* after the research director looked at premium summaries on partial data that included ~30 test runs (data/SPLITS.md, disclosure 3); its test number is therefore not a blind out-of-sample discovery. The rounds-2–12 variant is reported as a robustness check for that reason. Further disclosed exposures (data/SPLITS.md): a 24-test-run shuffle smoke test, an early all-run TF-IDF fit for textsim (since replaced by a dev-only fit), and cross-family agreement numbers computed on the mod-3 subset incl. test runs. The test evaluation is locked and run once, but not fully blind.

Limitations

- **Label = condition.** On llm_v3 all 70 `private` runs are judged y = 1 and `none`/`legit` runs are y = 0 by construction (no channel to judge), so y equals the condition in all 125 runs: the endpoint is effectively "channel granted and used", and detecting "agreement" and detecting "a channel was granted" coincide; the endpoint is coordination *communication*, not realised overcharge. A separate firm-framing stratum (`llm_v3fp`, `fp_stratum.py`, frozen dev thresholds) is an exploratory transfer check of "agreement vs. channel without agreement" — it changes framing, wording, refusals and price levels at the same time, so it cannot resolve the confound within llm_v3. The placebo stratum (`llm_v3plc`, same seeds and prompt, messages never delivered) addresses part of it within one generator: premium and auditor flags are no lower without message delivery, so delivered private coordination is not needed for what our conduct-based detectors pick up (n = 15, exploratory; equivalence not shown; focal opening prices and tacit public-price coordination are not separated).
- **Evasion not tested.** Sellers in the headline data were never told prices are watched. In the exploration stratum `llm_v3mon`, with an announced price monitor the judge found no agreements (0/14), so no evasive colluders were produced; robustness of the premium to evasion remains open.
- **The premium requires comparison firms.** As specified it is undefined in a two-firm market and cannot detect a uniform price increase shared by all four firms. That is where text or dependence statistics might be needed, and we did not test it.

- **Generator-specific premium.** Channel holders open higher before receiving any private message. Round 1 alone gives test run AUROC 0.90; rounds 2–12 give test 0.90 and dev 0.92. Removing observed round 1 does not estimate what would happen if sellers changed their opening strategy.
- Single seller model family (GPT, gpt-6-luna), single framing for the headline data, 12 rounds, 4 firms.
- Agreement labels from one LLM judge (gpt-6.1-sol, same family as the auditor; blind to prices); manually spot-checked on 3 dev chats + pilots.
- Zero-shot auditors with one prompt per view (0 tuning attempts on the six original views; the post-hoc hint view E56 is one data-informed revision); a tuned or few-shot auditor might use the stats table better.
- Small dev split (30 runs, base rate 2/3) → dev-fitted thresholds and fusion weight are noisy. The Youden rule breaks ties toward the higher threshold: conservative on FPR only — with perfect dev separation (premium) the cut sits at the lowest dev-positive score, which can cost test TPR.
- The sibling's information-theoretic detectors may have been developed while looking at llm_v3 runs that include our test runs; their test numbers are a comparator, not a locked held-out result.

What to do next

1. **Evasion-aware colluders.** The premium already discriminates strongly before any partner message is read; robustness to sellers that conceal this opening-price cue remains untested. Under our `llm_v3mon` announcement (prices monitored, flagged pairs lose half their earnings) gpt-6-luna produced no judge-labelled agreements (0/14), so testing evasion needs a red-teamed or instructed colluder (or a less safety-tuned seller model), including colluders that coordinate on gradual drift; we expect the premium and stats-table views to degrade first.
2. **Make text matter.** Text should become necessary when conduct alone is ambiguous: legit cooperation that moves prices, or collusion signalled in public through innocuous-looking wording (a supervised text model already finds such a trace here). Train/transfer that text model across seller models and framings before trusting it.
3. **Label ≠ condition.** Include placebo-matched negatives and test whether public text or conduct distinguishes delivered private coordination from an offered but undelivered channel, across seller models and games where a delivered agreement actually changes prices. Add a one-sided placebo (offer to one seller only) to separate offer-induced focal pricing from tacit coordination on public prices.
4. **Auditor design.** The stats-table views already include the premium. A one-sentence premium hint was tested post hoc (§Exploration, E56: mixed). Test it prospectively on a fresh split, ablate the premium column, evaluate any stats-first → LLM cascade on fresh data, and test tuned/few-shot auditors on a fresh split.
5. **Breadth.** Other seller model families, more firms/rounds, and a second market type.

Appendix A — LLM market, DEV split (runs 0000–0029; development only, superseded by the locked test above)

Headline numbers above come from the completed, single locked test analysis (`final_analysis.py --split test` via `finalize_test.sh`, runs 0030–0124). The development results below are retained for context. Numbers below are **dev** (runs 0000–0029: 20 positive, 10 negative = 5 `none` + 5 `legit`),

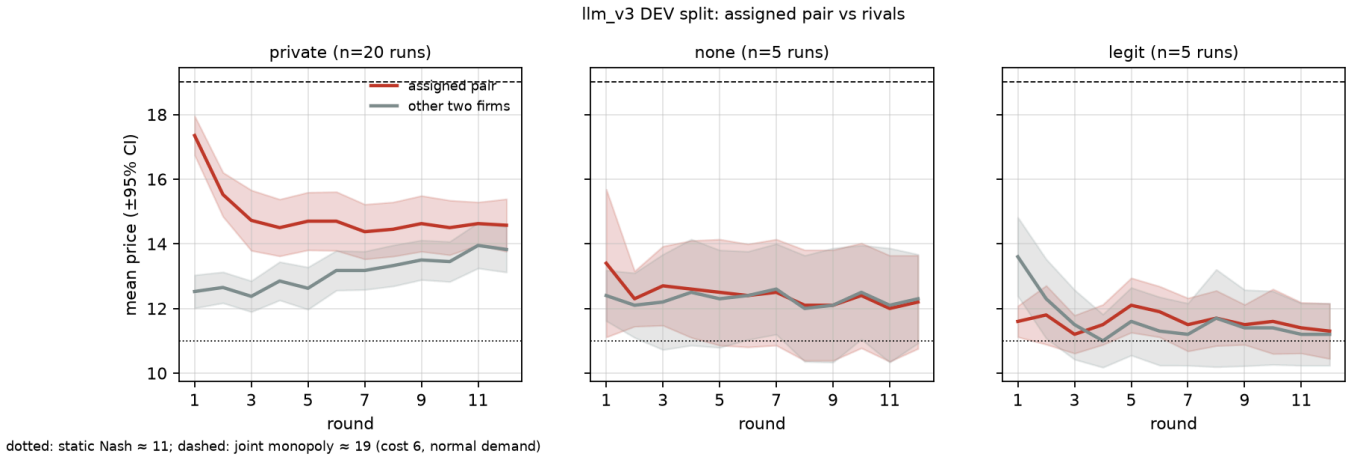
indicative only. `uv run python final_analysis.py --split dev` → `results/analysis/final_llm_v3_gpt-6.1-sol_dev.json`.

No explicit public coordination was flagged. `hardset.py` (keyword list + a cheap-tier LLM check of every run's public messages: gpt-6-luna for 94 runs, deepseek-v4.1-flash via llm.py's backend auto-order for 31) flagged **0 / 94** runs checked at that point (final: **0 / 125**, re-checked in the locked run) for explicit coordination language in public. Within the limits of a keyword list + one LLM check, the sellers keep coordination in the private channel, so the LLM set is effectively all "hard" set; an "easy" set does not occur naturally here.

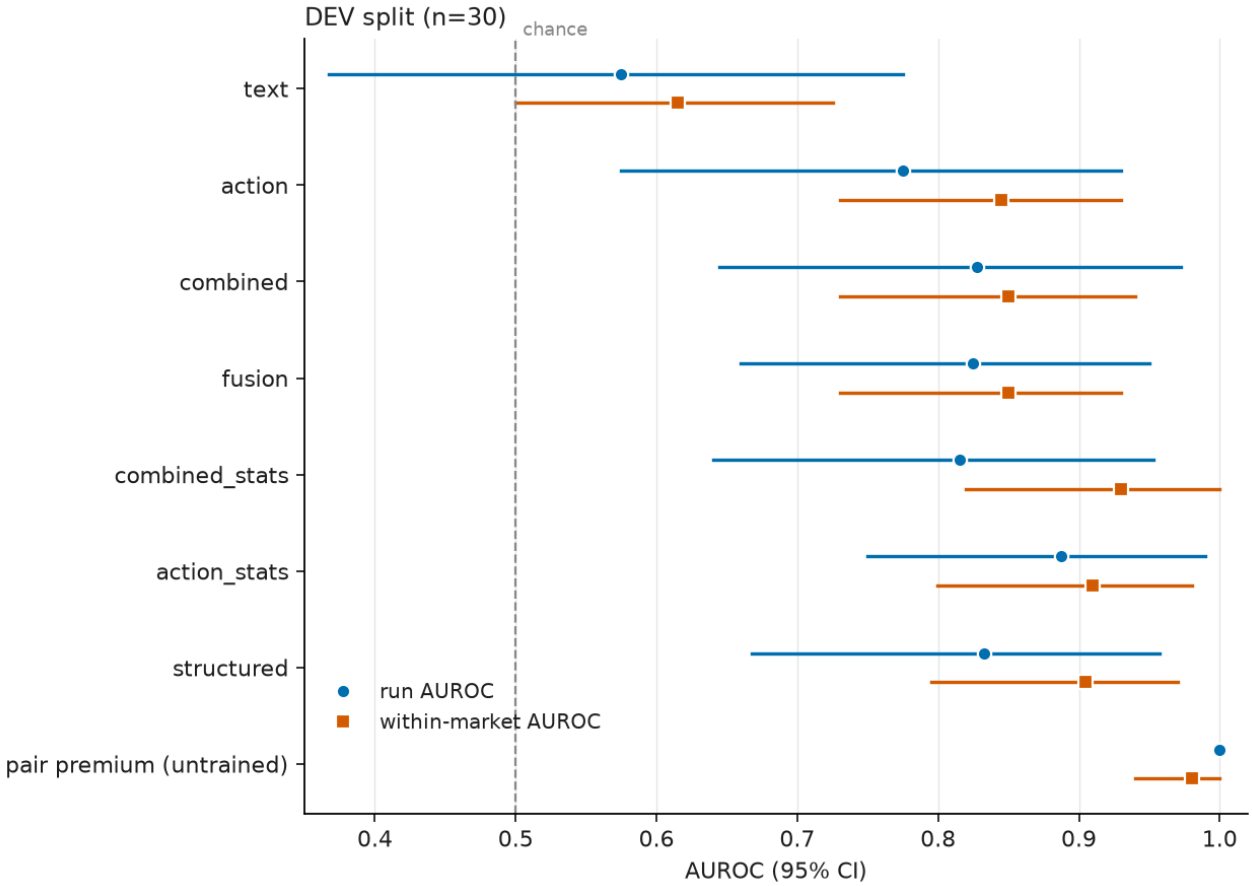
Scorers: six gpt-6.1-sol zero-shot auditor views; the **untrained pair premium** (pair mean price − mean of the other two firms, every pair, all rounds; run score = max over pairs; nothing fitted); a **late fusion** w·text + (1−w)·action of the unimodal auditor scores (w fitted on dev: 0.6). Dev, 2000 class-stratified run bootstraps; table copied verbatim from the generated `results/analysis/report_tables_dev.md` (flag rates at dev thresholds: see the flag-rate table there; y=1 / all negatives: text .25/.10, action .65/.10, action_stats .85/.10, combined .80/.20, combined_stats .90/.30, structured .65/.10, fusion .70/.10, premium 1.00/.00):

scorer	run AUROC (detection)	within-market AUROC (localisation)	pooled pair AUROC	pair-ID acc
text	0.57 [0.37, 0.78]	0.61 [0.50, 0.72]	0.56 [0.45, 0.68]	0.31 [0.17, 0.45]
action	0.78 [0.57, 0.93]	0.84 [0.73, 0.93]	0.82 [0.70, 0.92]	0.52 [0.30, 0.72]
action_stats	0.89 [0.75, 0.99]	0.91 [0.80, 0.98]	0.91 [0.80, 0.98]	0.75 [0.55, 0.90]
combined	0.83 [0.64, 0.97]	0.85 [0.73, 0.94]	0.80 [0.68, 0.90]	0.55 [0.35, 0.75]
combined_stats	0.81 [0.64, 0.95]	0.93 [0.82, 1.00]	0.89 [0.78, 0.97]	0.85 [0.70, 1.00]
structured	0.83 [0.67, 0.96]	0.91 [0.80, 0.97]	0.89 [0.78, 0.96]	0.70 [0.50, 0.90]
fusion	0.82 [0.66, 0.95]	0.85 [0.73, 0.93]	0.82 [0.70, 0.92]	0.52 [0.30, 0.72]
premium	1.00 [1.00, 1.00]	0.98 [0.94, 1.00]	0.99 [0.96, 1.00]	0.95 [0.85, 1.00]
premium_r2plus	0.92 [0.80, 1.00]	0.95 [0.89, 1.00]	0.96 [0.91, 0.99]	0.90 [0.75, 1.00]

(Thresholds are dev-fitted, so the flag rates on dev are optimistic; honest test FPR is in the headline section.) Why the premium is so strong: private-channel sellers choose their opening price and first private message in the same call, before reading a partner message (see §Exploration "When does the premium appear?"; earlier versions wrongly said the pair "agrees in round 1"), while the rivals open at ≈ 12. In three inspected dev chats (0005, 0012, 0016), sellers lower their opening prices after rivals undercut them; this does not establish that coordination ends. The opening-round gap alone carries much of the signal; without round 1 the premium still matches or beats every original LLM view on dev, but no longer significantly vs structured/action_stats (premium_r2plus − structured: run +0.09 [−0.10, +0.29]; it still beats text, and action/combined on localisation). The LLM auditors also see round 1.



Descriptive (dev, `uv run python fig_mechanism.py --split dev` → `logs/mechanism_dev.txt`): private pairs open near 17–18 vs ≈ 12.5 for the rivals, fall back within ~3 rounds, but on average stay above the rivals to the end: late-window (rounds 9–12) pair-minus-rivals premium +0.90 (none −0.07, legit +0.15), positive in 12/20 private runs. So the signal is a large round-1 gap plus a smaller persistent gap that roughly 40% of private pairs lose. **Label ≈ condition:** all 20 dev `private` runs are labelled y = 1 (19 explicit, 1 info-exchange), so on this generator "agreement" and "had a channel" coincide; see Limitations. Paired contrasts (dev, from the generated file): Δ_hybrid[combined] run +0.052 [−0.123, +0.210], within +0.005 [−0.050, +0.045]; Δ_hybrid[combined_stats] within +0.085 [+0.040, +0.135]; combined − fusion run +0.003 [−0.120, +0.120]; action_stats − action within +0.065 [+0.020, +0.110]; combined_stats − action_stats within +0.020 [+0.000, +0.050]; premium − structured run +0.168 [+0.043, +0.333], within +0.075 [−0.010, +0.185]; premium − action_stats run +0.113 [+0.010, +0.250].



Development interpretation (compare the completed test and replication results above): on dev, adding the public text to the actions shows no resolved gain for the LLM auditor (combined − action within +0.005 [−0.05, +0.045]; joint reading ≈ late fusion of the two unimodal auditors). A plausible reason, not tested causally, is that the sellers keep coordination private and their public messages are bland. The views that include a *computed pair-statistics table* (action_stats, combined_stats, structured) localise better than those without; separately, a single untrained statistic (the pair premium) beats every original LLM view on dev. (Whether the table's benefit is the premium row specifically is not ablated.) This fits the sibling project's finding that dependence statistics detect coordinated *variation*, not *level*: these colluders coordinate on level, and the premium measures level directly.

Information-theoretic dependence detectors (sibling Q1.4, dev). Using the sibling project's per-pair scores on the same runs (`../collusion-14-infotheory/results/eval/llm_v3_ext.json`, read-only; block `external_infotheory` in the final_analysis output): run AUROC pCMI 0.47 [0.22, 0.72], MI 0.60 [0.40, 0.80], TE 0.59 [0.32, 0.83], Pearson 0.21 [0.06, 0.39] (below chance; hypothesis, not tested: colluding pairs hold prices flat, so co-vary less); within-market MI 0.85, Pearson 0.76, TE 0.67, pCMI 0.55. Premium − pCMI run AUROC +0.53 [+0.27, +0.78]. Their independently coded `pair_level` statistic matches our premium's aggregate metrics (run 1.00, within 0.98); per-pair equality not checked. So on this generator, dependence-in-variation statistics are weak, level statistics are strong.

Illustrative differences in pair scores (gallery, `results/analysis/gallery_dev.md`; both approaches clear this run at their frozen thresholds, so this illustrates differing rationales, not better classification): in `legit` run 0001 the partnership pair has a positive premium (+0.62 all-round) and is the premium's top pair, while the text/structured auditors read the public partnership notice and give that pair 4–7/100. On dev the premium's run-level threshold happens to sit above it, but this is the failure mode a price-only screen has with legitimate cooperation.

Evidence accuracy (dev, `evidence.py` → `results/analysis/evidence_llm_v3_gpt-6.1-sol_dev.json`; 5 views, combined_stats not checked). Syntactic checks only (cited round/firm exist; extracted price numbers vs the log), not whether the evidence supports the claim. All cited rounds/firms are valid. In the four action-bearing views 45–55% of items contain a price claim; of these, 51–65% match the cited firm's price in the exact cited round, 81–92% within ±1 round, 95–100% match some firm's price nearby (text view: only 3 price claims). On positive runs 47–59% of cited firms belong to the colluding pair (chance 50%): auditors cite market-wide observations, so their evidence rarely singles out the colluders. No verbatim quotes were cited.

Audit cost (prompt size per call; `uv run python cost.py` → `results/analysis/cost_llm_v3.json`). text 9.2k chars, action 3.3k, action_stats 3.7k, combined 11.4k, combined_stats 11.8k, structured 12.0k (≈ chars/4 tokens); one gpt-6.1-sol call per run per view. The premium requires no LLM calls; a premium-first triage that sends only the top 75% of dev runs to the LLM audit keeps all positives (dev base rate is 2/3; test: auditing the top 50% / 75% sends 0.84 / 1.00 of y=1 runs, see `report_tables_test.md`). End to end (run sent by the premium screen AND flagged by the LLM view at its dev threshold), the cascade cannot exceed its premium screen's recall but can cut false positives: dev, auditing all runs, action_stats catches 0.85 of y=1 runs at FPR 0.10 and names the correct pair in 0.65; structured 0.65 / 0.10 / 0.55 (`triage.*.cascade_*` in the final json). A proposed architecture is premium screen → LLM audit for explanation/evidence and false-positive filtering (explanation usefulness itself is not measured here). **Does any LLM view add to the premium? (`incremental.py`, iter 7, dev.)** An untrained, label-free combination (average percentile rank of premium and the view) never beats the premium alone on dev: combo − premium run AUROC from −0.10 [−0.24, −0.005] (text) to −0.015 [−0.065, 0.00] (action_stats); within-market ≤ 0. On y=1 runs, every view gets the colluding pair right while the premium gets it wrong in at most 1 of 20 runs. Dev is at the premium's ceiling, so this can only show harm there; the completed test comparison is in Findings, item 5.

Caveat (22:00 codex review, `logs/review_interim.md`): `legit` runs were initially audited WITHOUT their public partnership notice. Fixed 22:10 (auditors.py regenerates `public_info`; legit audits invalidated and re-run; cached audits are now checked against the current prompt hash/length and re-run if stale).

Supervised non-LLM baselines (dev; `baselines.py --ds llm_v3 --split dev`, LR, 5-fold CV grouped by run inside dev, n = 30). Run AUROC text 0.79 [0.61, 0.93], action 0.96 [0.86, 1.00], combined 0.96, structured 0.95; pair-ID accuracy text 0.90, action 0.95 (action = computed price/outcome features; supervised CV inside dev, labels of other dev runs). Hybrid Δ ≈ 0 (ceiling). A *trained* classifier on price features far outperforms the zero-shot LLM reading the same prices (not like-for-like: the LR sees in-distribution labels). The text LR beats the LLM text auditor by a wide margin (pair-ID 0.90 vs 0.31). An informal look at its top-weighted n-grams during development (no saved artifact) suggested margin-flavoured phrasing for positives ("considered price", "sustainable") and the `legit` pairs' "shipping partnership" notices for negatives.

Untrained public-text convergence (new, iter 6, dev; `textsim.py`). Hypothesis: if partners' private coordination shapes their public wording, the two colluders' public messages should resemble each other more than other pairs' do. Score = TF-IDF cosine of the pair's public messages − mean cosine of the other pairs (vectorizer fit on dev text only, no labels; nothing tuned). Dev: within-market AUROC **0.81 [0.69, 0.91]** vs the zero-shot LLM text auditor 0.61 (paired Δ +0.20 [+0.07, +0.31]); pair-ID 0.55. It showed no detection ability on dev (run AUROC 0.48 [0.24, 0.72], inconclusive). So the public text does contain a localisation cue that the zero-shot LLM text auditor does not reflect in its pair scores. That the cue comes from the private agreement leaking into public wording is a plausible mechanism, not shown (label ≈ condition here). Pre-registered before this scorer was evaluated on test (earlier disclosed test exposures of other scorers still apply, `data/SPLITS.md`).

Appendix B — Scripted mechanism check (non-LLM baselines, `baselines.py`, 5-fold CV grouped by run, 320 runs)

`uv run python baselines.py --ds scripted_v2 --learner lr` → `results/baselines/scripted_v2_lr.json`.

- scripted_v1 (strong collusion): action features alone reach AUROC 1.0 → Δ_hybrid = 0 by ceiling; uninformative.
- scripted_v2 (weak, low-markup collusion; hard set = 280 runs excl. explicit public talk), logistic regression on nested feature sets, run-level AUROC [95% bootstrap CI]:

view	**intact**	**messages shuffled (same label+condition donor, same CV test fold)**
text (TF-IDF)	0.839 [0.788, 0.882]	0.837 [0.787, 0.882]
action stats	0.951 [0.922, 0.973]	0.951 (unchanged by construction)
combined raw (text + action)	0.961 [0.939, 0.979]	0.961 [0.940, 0.979]
structured (action + say-vs-do consistency)	1.000 [1.000, 1.000]	0.748 [0.680, 0.811]

  Δ_structured = +0.049 [0.027, 0.079]; Δ_combined = +0.010 [0.004, 0.019]. Shuffling leaves both single modalities and the raw union untouched but collapses the structured model → its gain comes from the say-vs-do alignment features, i.e. the mechanism the templates were built to contain.

- Methodological cautions found on the way: (i) mixing learners (boosting for numeric, LR for text) produced an apparent shuffle collapse on scripted_v1 that vanished with identical learners; (ii) drawing shuffle donors from the whole dataset let a boosted text model recognise memorised training-fold text (text-only AUROC *rose* 0.91→0.995 under shuffle; historical debugging value, EXPERIMENTS.tsv E05, configuration not retained); donors are now drawn within the CV test fold.

Appendix C — Additional test-split detail (locked test split, n = 95; moved from §Results for length)

Flag rates at the dev-chosen thresholds (test, honest FPR):

stratum	n	text	action	action_stats	combined	combined_stats	structured	fusion	premium	premium_r2plus
all_negatives	45	0.11 [0.05, 0.23]	0.04 [0.01, 0.15]	0.04 [0.01, 0.15]	0.20 [0.11, 0.34]	0.18 [0.09, 0.31]	0.04 [0.01, 0.15]	0.09 [0.04, 0.21]	0.16 [0.08, 0.29]	0.20 [0.11, 0.34]
legit	15	0.07 [0.01, 0.30]	0.00 [0.00, 0.20]	0.07 [0.01, 0.30]	0.27 [0.11, 0.52]	0.20 [0.07, 0.45]	0.00 [0.00, 0.20]	0.07 [0.01, 0.30]	0.33 [0.15, 0.58]	0.33 [0.15, 0.58]
none_all	30	0.13 [0.05, 0.30]	0.07 [0.02, 0.21]	0.03 [0.01, 0.17]	0.17 [0.07, 0.34]	0.17 [0.07, 0.34]	0.07 [0.02, 0.21]	0.10 [0.04, 0.26]	0.07 [0.02, 0.21]	0.13 [0.05, 0.30]
none_highmarkup_proxy	1	1.00 [0.21, 1.00]	0.00 [0.00, 0.79]	0.00 [0.00, 0.79]	1.00 [0.21, 1.00]	1.00 [0.21, 1.00]	1.00 [0.21, 1.00]	1.00 [0.21, 1.00]	0.00 [0.00, 0.79]	0.00 [0.00, 0.79]
none_lowmarkup	29	0.10 [0.04, 0.26]	0.07 [0.02, 0.22]	0.03 [0.01, 0.17]	0.14 [0.06, 0.31]	0.14 [0.06, 0.31]	0.03 [0.01, 0.17]	0.07 [0.02, 0.22]	0.07 [0.02, 0.22]	0.14 [0.06, 0.31]
private_y1	50	0.10 [0.04, 0.21]	0.20 [0.11, 0.33]	0.72 [0.58, 0.82]	0.52 [0.39, 0.65]	0.68 [0.54, 0.79]	0.40 [0.28, 0.54]	0.28 [0.17, 0.42]	0.92 [0.81, 0.97]	0.84 [0.71, 0.92]

Label≈condition check: detection (flag rate) among y=1 runs by realised late-window relative premium of the colluding pair:

stratum	n	text	action	action_stats	combined	combined_stats	structured	fusion	premium	premium_r2plus
late_rel_premium_le_0.5	20	0.05 [0.01, 0.24]	0.25 [0.11, 0.47]	0.65 [0.43, 0.82]	0.55 [0.34, 0.74]	0.60 [0.39, 0.78]	0.35 [0.18, 0.57]	0.35 [0.18, 0.57]	0.80 [0.58, 0.92]	0.60 [0.39, 0.78]
late_rel_premium_gt_0.5	30	0.13 [0.05, 0.30]	0.17 [0.07, 0.34]	0.77 [0.59, 0.88]	0.50 [0.33, 0.67]	0.73 [0.56, 0.86]	0.43 [0.27, 0.61]	0.23 [0.12, 0.41]	1.00 [0.89, 1.00]	1.00 [0.89, 1.00]

Other test results (all in `results/analysis/report_tables_test.md`): sibling information-theoretic detectors are weak (run AUROC pCMI 0.54, MI 0.62, TE 0.45, Pearson 0.56; their `pair_level` = our premium, 0.95); premium-first cascade end to end (premium sends top 50% of runs, LLM view must also flag): action_stats recall 0.60 / FPR 0.02 / correct pair 0.44, structured 0.34 / 0.00 / 0.26 — relative to its upstream premium screen (top 50%: recall 0.84, FPR 6/45 = 0.13) the AND-gate trades recall for fewer false positives; which is better depends on the costs of misses, false alarms and audit calls; evidence items name a colluding firm 54–62% of the time on y=1 runs (chance 50%); 81–93% of price claims pass a permissive regex check (numbers match the log within ±1 round; exact-round 49–63%; action-bearing views; reasoning itself not validated). Cross-family (Claude Sonnet 5.5 vs gpt-6.1-sol, all 42 mod-3 runs, 24 y=1, dev+test pooled; `cross_family.py` → `results/analysis/llm_v3_crossfamily_claude-sonnet-5-5.json`): the **main pattern replicates across model families** — text lowest (run AUROC GPT 0.51 / Claude 0.58), action 0.76 / 0.80, combined 0.74 / 0.74; stats views (action_stats 0.85 / 0.91, structured 0.79 / 0.90, combined_stats 0.76 / 0.92) are highest for Claude, while for GPT only action_stats clearly is (point estimates, no CIs, n = 42, dev + test pooled). Adding text to action has no resolved effect for either auditor (Δ pair AUROC combined − best single: GPT +0.01 [−0.05, +0.08], Claude −0.01 [−0.10, +0.07]); adding stats does (action_stats: GPT +0.09 [+0.02, +0.16], Claude +0.09 [+0.00, +0.19]). Per-pair agreement between the two auditors is only moderate (Spearman 0.31 text … 0.59 structured; top-pair agreement 0.38–0.71), so individual verdicts are model-dependent even though the tested aggregate comparisons agree.

Finding 4 in full (legitimate cooperation at matched recall; moved from §Results):
Legitimate cooperation: no LLM advantage at matched recall. At dev-chosen thresholds the premium flags 5/15 (0.33) `legit` partnership runs, while structured and action flag 0/15 (Wilson CI [0, 0.20]); these views can read the public partnership notice (no notice-removal ablation was run). But they operate at much lower recall (0.40 and 0.20 of y=1 runs vs the premium's 0.92), and the text-reading combined views flag legit runs about as often as the premium (combined 0.27, combined_stats 0.20). Comparing at a common *minimum recall target* (post-lock, descriptive; `uv run python matched_recall.py` → `results/analysis/matched_recall_llm_v3_test.json`; oracle thresholds read off the test positives; ties can push a scorer's recall above the target) gives mixed results at moderate recall and a premium advantage at high recall. Legit runs flagged: target 0.5 — premium 0/15 (TPR 0.50), structured 0/15 (0.54), action_stats 1/15 (0.50); target 0.7 — premium 2/15 (0.72), action_stats 1/15 (0.72), structured 3/15 (0.78); target 0.9 — premium 5/15 (0.90), action_stats 9/15 (0.92), structured 11/15 (0.92); `none` runs at target 0.9: 2/30 vs 20/30 and 19/30. At these recall targets the LLM views show no consistent advantage over the premium in clearing legitimate runs (n = 15 legit runs; no CIs). The contribution of the public partnership notice was not isolated. Numerical note: the frozen and oracle premium cuts both represent 2/3 but differ in floating-point rounding (premium scores of mathematically tied runs differ at ~1e-15), which explains a one-run recall difference (46/50 vs 45/50); it has no substantive meaning.

Text-classifier inspection in full (moved from §Results): Post-headline inspection (`uv run python text_lr_inspect.py` → `results/analysis/text_lr_inspect.json`; descriptive all-data refit, not an evaluation; it does not show which features drive the out-of-fold performance): top positive uni/bigrams are bland margin phrasing ("considered price", "supports reliable", "balances customer"), top negative ones competitive phrasing ("price competitive", "your business", "keeping"). At the firm level (all 125 runs; each firm's concatenated public messages; role = colluding firm / rival firm in the same private run / firm in a `none` run): "considered price" 25/140 colluders, 11/140 rivals, 0/140 `none` (0/80 `legit`) — so rivals pick it up too (condition-level shift); but the **pair-specific cue is large**, and mostly it is what colluders *stop* saying: "keeping" 67/140 vs 133/140 vs 139/140, "business" 6/140 vs 45/140 vs 56/140, while "adjusting our" rises (51/140 vs 12/140 vs 21/140). This is consistent with the text LR's perfect pair-ID (1.00 vs premium 0.89). The message shuffle (donors share label + condition) cannot remove a condition-level style signal, so its

shuffle-robust 0.97 is not evidence against it. Interpretation (not tested): public text carries an *indirect* trace — colluders drop competitive language — that a supervised model trained in-distribution learns but a zero-shot LLM auditor reading for explicit coordination does not use. Whether it reflects the channel prompt, the communication or the conduct cannot be separated here (label = condition). Whether it transfers to other seller models or framings is untested.

Appendix D — Exploration strata detail (moved from §Results for length; text unchanged)

Firm framing, channel offered, no agreement found (`llm_v3fp`, 16 `private` runs, `uv run python fp_stratum.py --ds llm_v3fp` → `results/analysis/fp_stratum_llm_v3fp.json`). Under the "pricing manager + competition authority" framing, the judge finds **no agreement in 16/16 runs**: sellers given a private channel declined to coordinate (replicating the pilot). These are natural "channel, no agreement" negatives, so a scorer that tracks agreement rather than framing should rarely flag them. Flag rates at the dev thresholds [Wilson 95%]: text 0.06 [0.01, 0.28], action 0.06 [0.01, 0.28], structured 0.25 [0.10, 0.50], premium 0.44 [0.23, 0.67], combined 0.50, action_stats 0.50, combined_stats 0.63 [0.39, 0.82] — vs 0.03–0.17 on lab-framing `none` test runs. Mean prices are close to the lab `none` runs (13.2 in round 1, 12.5 in round 12, vs 12.9 / 12.0; `uv run python fp_prices.py` → `results/analysis/fp_prices.json`), but the source of the false alarms was not established (pair-level price differences under the other framing are a guess, not analysed). **Thresholds and stats-based scorers did not transfer to a framing shift**; the stats table, which helped on llm_v3, raised false alarms here. With only y = 0 runs, no AUROC can be computed. Exploratory (n = 16, one framing change that also changes wording and refusals).

Announced price monitor (`llm_v3mon`, lab framing + public announcement that prices are monitored; 14 private, 6 none; the note says an automated monitor reviews prices + public board, cannot see private messages, and halves the earnings of flagged pairs — announced only, no monitor is run). `uv run python deterrence.py` → `results/analysis/deterrence.json`. The pre-registered question (does the premium survive colluders who know prices are watched?) **could not be tested: the monitored sample contains no judge-labelled coordination communication.** The judge finds an agreement in **0/14** monitored private runs [Wilson 0.00, 0.22] vs **70/70** unmonitored lab private runs [0.95, 1.00] (Fisher $p \approx 3 \cdot 10^{-16}$). The cohorts are unpaired (seeds 3000000–3000124 vs 7000000–7000019; 70 vs 14 private, 35 vs 6 `none` runs); same seller model (gpt-6-luna), 12 rounds, and system prompts that differ only by the monitor note. Lexical indicators (regexes, not validated measures): digit matches in 0/336 monitored private messages vs 1488/1680 (88.6%) on llm_v3; independence/refusal phrases (e.g. "I can't coordinate prices") in 335/336 (99.7%) vs 613/1680 (36.5%). Mean markup over cost in monitored private runs is 6.16 vs 6.27 in monitored `none` runs (llm_v3: private 7.66, none 5.84); assigned-pair premium +0.23 (llm_v3 private +1.65); premium > 1 in 1/14 runs (llm_v3 61%) — close to no-channel levels, though not shown equivalent. So on this sample, announcing a monitor coincides with no coordination talk rather than with hidden coordination. Caveats: n = 14, one seller model, one judge whose reliability on refusal-style chats was not independently checked; we cannot distinguish safety-triggered refusal from strategic deterrence, nor exclude tacit coordination, and it says nothing about how a model that does try to evade would look. **False alarms on this stratum** (all 20 runs are y = 0; frozen dev thresholds; `uv run python monitor_stratum.py --ds llm_v3mon` → `results/analysis/monitor_stratum_llm_v3mon.json`), flag rate on the 14 private-no-agreement / 6 `none` runs [Wilson 95%]: text 0.00 [0.00, 0.22] / 0.00 [0.00, 0.39], action 0.21 / 0.50, structured 0.21 / 0.17, premium 0.36 [0.16, 0.61] / 0.50 [0.19, 0.81], action_stats 0.43 / 0.33, combined 0.43 / 0.33, combined_stats 0.36 / 0.33 — vs 0.03–0.17 on lab `none` test runs. Like the firm-framing stratum, a prompt change after which no agreements are found leaves the premium and the stats/combined

views flagging a third to a half of runs: the premium's run score is the largest premium over all six pairs, and in 8/20 monitored runs it reaches the frozen cut of 0.67 price units (max 2.29), more often than in lab `none` test runs (why was not analysed). Exploratory, n = 20.

Round-1 citations by the LLM auditors (post-lock, descriptive; `uv run python evidence_rounds.py --ds llm_v3 --split test` → `results/analysis/evidence_rounds_llm_v3_test.json`; `--ds llm_v3rep --split all` for the replication). Each audit cites up to 4 evidence items, each tied to a round. On the 95 test runs, round 1 accounts for 4.5–10% of cited items depending on the view (text 6.8%, action 4.5%, combined 5.0%, combined_stats 8.5%, action_stats 10.1% [7.7, 12.7], structured 9.0%). Uniformly cited rounds would give 8.3%. Among audits of y = 1 runs, round 1 is cited at all in 16–24% of text/action/combined audits and 42–48% of the stats-table views (11–31% of y = 0 audits). In the replication, round 1 accounts for 5.8–11.8% of cited items. The views without a stats table cite round 1 at or below the uniform rate, although round 1 is where the pair premium is already large. The headline run score (`collusion_present`) correlates with both the round-1 and the rounds-2..12 premium (test Spearman: action_stats 0.52 vs 0.58; combined 0.42 vs 0.40). These are citation frequencies and descriptive associations. They do not show which rounds drove a verdict, nor do they explain the stats-table benefit.

Telling the auditor where to look (`action_stats_hint`; post-lock exploration E56; `uv run python auditors.py --ds llm_v3 --views action_stats_hint` (+ `--ds llm_v3rep`), then `uv run python hint_view.py` → `results/analysis/hint_view.json`). This view is `action_stats` plus one fixed sentence. The sentence says that colluding pairs typically price above the other two firms, often from round 1, and asks the auditor to check each pair's premium and round-1 prices. We wrote it once, after the locked test and the replication, using what those analyses had shown. No further prompt search followed this data-informed revision. Because the hint encodes knowledge taken from the test and replication data, neither split is held out for it. Paired, class-stratified bootstrap over runs:

- **Test (n = 95).** Run AUROC rises from 0.86 to 0.93 (Δ +0.07 [−0.00, +0.15]). Within-market AUROC rises from 0.89 to 0.96 (Δ +0.06 [+0.02, +0.11]). Pair-ID rises from 0.66 to 0.83. The remaining gap to the premium is +0.02 [−0.02, +0.07] (run) and +0.00 [−0.03, +0.03] (within-market).
- **Replication (n = 30).** No resolved run-level gain: 0.796 vs 0.798, Δ −0.002 [−0.18, +0.16]. Within-market AUROC is 0.99 for both, near ceiling. The premium's point estimate remains higher at 0.92 (Δ +0.12 [−0.03, +0.28]).
- **Dev (n = 30).** Run AUROC 0.93 vs 0.89; within-market 0.98 vs 0.91. Neither change is resolved.

The hint makes the run score track the premium more closely: Spearman with the max-pair premium rises from 0.60 to 0.87 on test and from 0.65 to 0.75 on the replication. Pointing the auditor at the premium can therefore recover much of the premium's ranking on test, but the gain does not show on the 30 replication runs (where action_stats' pair scores already localised near ceiling). The test gaps to the premium are unresolved; this establishes neither equivalence nor superiority. A cheaper design is to compute the statistic and ask the LLM only for explanation.

Appendix E — Reproduce

The experiments are complete. For a read-only offline check of the shipped artifacts, run `python3 verify_artifacts.py` (standard library only; no LLM calls, metric recomputation, or writes). The post-hoc manifest and completion receipt were added in iteration 19 to preserve the handoff; they are not pre-registration evidence. See README.md for artifact locations and safe rendering commands. The generation/audit commands below document how the existing results were produced; the two completed run-once entry points now refuse to rerun even in a fresh clone.

```
uv run python llm_market.py --ds llm_v3 --plan private:70,none:35,legit:20 --rounds 12 --framing lab --games-
parallel 2 --seed 3000
uv run python label_llm.py --ds llm_v3
uv run python auditors.py --ds llm_v3 --variant orig   # default = all six views
uv run python hardset.py --ds llm_v3
uv run python final_analysis.py --split dev    # development only; the test headline runs ONLY via finalize_test.sh
uv run python evidence.py --split test         # evidence accuracy + prompt cost
uv run python auditors.py --ds llm_v3 --variant shuffle --shuffle-seed 0 --views text,combined,structured   # seeds
0,1,2
uv run python baselines.py --ds llm_v3 --learner lr --split test   # CV inside the split
uv run python textsim.py --split test           # untrained public-text convergence
uv run python report_tables.py --split test     # renders all REPORT tables from results/*.json
uv run python fig_mechanism.py --split all      # descriptive price paths by condition (no metric)
uv run python fig_roc.py                        # test ROC + dev operating points (descriptive)
uv run python fig_summary.py                    # Figure 1 summary (reads existing results JSON only)
uv run python text_lr_inspect.py                # n-gram / firm-level text cue inspection (descriptive)
uv run python incremental.py --split test       # premium + LLM view rank-average (run by finalize_test.sh)
uv run python fp_stratum.py --ds llm_v3fp       # firm-framing exploration stratum
uv run python fp_prices.py                      # firm-framing vs lab none mean prices (descriptive)
uv run python monitor_stratum.py --ds llm_v3mon # announced-monitor stratum (frozen thresholds)
uv run python deterrence.py                     # agreement / message / markup comparison llm_v3 vs llm_v3mon
uv run python matched_recall.py                 # legit/none flags at matched recall (post-lock, descriptive)
uv run python round1_effect.py --ds llm_v3      # round-1 (pre-message) prices by role (descriptive)
uv run python early_detection.py --ds llm_v3 && uv run python fig_early.py   # premium AUROC vs rounds observed
uv run python evidence_rounds.py --ds llm_v3 --split test   # do auditors cite round 1? (descriptive)
uv run python auditors.py --ds llm_v3 --views action_stats_hint && uv run python auditors.py --ds llm_v3rep --views
action_stats_hint
uv run python hint_view.py                      # premium-hint auditor exploration (E56, post-lock)
uv run python llm_market.py --ds llm_v3plc --match-ds llm_v3rep --rounds 12 --framing lab   # placebo channel (E58)
uv run python label_llm.py --ds llm_v3plc && uv run python auditors.py --ds llm_v3plc && uv run python placebo.py
uv run python replication.py                    # historical run-once command; now guarded by committed completion
receipt
./finalize_test.sh                              # historical locked-test command; now guarded by committed
completion receipt
uv run python auditors.py --ds llm_v3 --auditor claude-sonnet-5-5 --subset mod3   # cross-family subset
uv run python cross_family.py --ds llm_v3 --other claude-sonnet-5-5               # cross-family comparison
uv run python sim.py --dataset scripted_v2 --n-per-cond 40 --rounds 30 --seed 1 --levels 12,14 --strength 0.15,0.5
uv run python baselines.py --ds scripted_v2 --learner lr
```